Name
Date
Course

Adding and Subtracting Algebraic Fractions

Pre-Algebra Review

① $\frac{2x}{3}+\frac{x}{6}$

$\frac{4x}{6}+\frac{x}{6}$

$\boxed{\frac{5x}{6}}$

② $-\frac{5y}{6}+\frac{3y}{4}$

$\frac{-10y}{12}+\frac{9y}{12}$

$\boxed{\frac{-y}{12}}$

③ $\frac{z}{6}-\frac{2z}{7}$

$\frac{7z}{42}-\frac{12z}{42}$

$\boxed{\frac{-5z}{42}}$

④ $\frac{3x}{4}+\left(-\frac{3x}{7}\right)$

$\frac{21x}{28}-\frac{12x}{28}$

$\boxed{\frac{9x}{28}}$

⑤ $\frac{y}{16}-\frac{5y}{4}$

$\frac{y}{16}-\frac{20y}{16}$

$\boxed{\frac{-19y}{16}}$

⑥ $-\frac{z}{8}+\frac{5z}{12}$

$\frac{-3z}{24}+\frac{10z}{24}$

$\boxed{\frac{7z}{24}}$

Beginning Algebra

⑦ $\frac{2}{x-6}+\frac{1}{x^2-2x-24}$

$\frac{2}{x-6}+\frac{1}{(x+4)(x-6)}$

$\frac{(x+4)\,2}{(x+4)(x-6)}+\frac{1}{(x+4)(x-6)}$

$\frac{2x+8}{(x+4)(x-6)}+\frac{1}{(x+4)(x-6)}$

$\boxed{\frac{2x+9}{(x+4)(x-6)}}$

⑧ $\frac{1}{3b}-\frac{4b}{3b-3}$

$\frac{1}{3b}-\frac{4b}{3(b-1)}$

$\frac{(b-1)\,1}{(b-1)\,3b}-\frac{4b\,(b)}{3(b-1)\,(b)}$

$\frac{b-1}{3b(b-1)}-\frac{4b^2}{3b(b-1)}$

$\boxed{\frac{-4b^2+b-1}{3b(b-1)}}$

⑨ $\frac{2}{a+3}+\frac{4}{a-5}$

$\frac{(a-5)\,2}{(a-5)(a+3)}+\frac{4\,(a+3)}{(a-5)(a+3)}$

$\frac{2a-10}{(a-5)(a+3)}+\frac{4a+12}{(a-5)(a+3)}$

$\boxed{\frac{6a+2}{(a-5)(a+3)}}$

⑩ $\frac{4}{x^2+3x-18}+\frac{1}{x-3}$

$\frac{4}{(x+6)(x-3)}+\frac{1}{x-3}$

$\frac{4}{(x+6)(x-3)}+\frac{1\,(x+6)}{(x-3)(x+6)}$

$\frac{4}{(x+6)(x-3)}+\frac{x+6}{(x-3)(x+6)}$

$\boxed{\frac{x+10}{(x+6)(x-3)}}$

⑪ $\frac{3c}{10c-4}+\frac{1}{2c}$

$\frac{3c}{2(5c-2)}+\frac{1}{2c}$

$\frac{(c)\,3c}{(c)\,2(5c-2)}+\frac{1\,(5c-2)}{2c\,(5c-2)}$

$\frac{3c^2}{2c(5c-2)}+\frac{5c-2}{2c(5c-2)}$

$\boxed{\frac{3c^2+5c-2}{2c(5c-2)}}$

⑫ $\frac{2}{a+1}+\frac{1}{a-2}$

$\frac{(a-2)\,2}{(a-2)\,(a+1)}+\frac{1\,(a+1)}{(a-2)(a+1)}$

$\frac{2a-4}{(a-2)(a+1)}+\frac{a+1}{(a-2)(a+1)}$

$\boxed{\frac{3a-3}{(a-2)(a+1)}}$

⑬ $\frac{3}{y+7}-\frac{4}{y}$

$\frac{(y)\,3}{(y)\,y+7}-\frac{4\,(y+7)}{y\,(y+7)}$

$\frac{3y}{y(y+7)}-\frac{4y+28}{y(y+7)}$

$\frac{3y-4y-28}{y(y+7)}=\boxed{\frac{-y-28}{y(y+7)}}$

⑭ $\frac{5z}{z^2+3z-10}+\frac{1}{z-2}$

$\frac{5z}{(z+5)(z-2)}+\frac{1\,(z+5)}{(z-2)(z+5)}$

$\frac{5z}{(z+5)(z-2)}+\frac{z+5}{(z-2)(z+5)}$

$\boxed{\frac{6z+5}{(z+5)(z-2)}}$

(15) $-\dfrac{1}{x^2-6x-16}+\dfrac{2}{x^2-3x-40}$

$-\dfrac{1}{(x-8)(x+2)}+\dfrac{2}{(x-8)(x+5)}$

$\dfrac{-1}{(x-8)(x+2)}\dfrac{(x+5)}{(x+5)}+\dfrac{2}{(x-8)(x+5)}\dfrac{(x+2)}{(x+2)}$

$\dfrac{-x-5}{(x-8)(x+2)(x+5)}+\dfrac{2x+4}{(x-8)(x+5)(x+2)}$

$\boxed{\dfrac{x-1}{(x-8)(x+2)(x+5)}}$

(16) $\dfrac{2}{y+5}-\dfrac{3}{2y}$

$\dfrac{(2y)}{(2y)}\dfrac{2}{(y+5)}-\dfrac{3(y+5)}{2y(y+5)}$

$\dfrac{4y}{2y(y+5)}-\dfrac{3y+15}{2y(y+5)}$

$\boxed{\dfrac{y-15}{2y(y+5)}}$

(17) $\dfrac{3z}{z^2-4z-5}+\dfrac{2}{z-5}$

$\dfrac{3z}{(z+1)(z-5)}+\dfrac{2}{z-5}$

$\dfrac{3z}{(z+1)(z-5)}+\dfrac{2}{(z-5)}\dfrac{(z+1)}{(z+1)}$

$\dfrac{3z}{(z+1)(z-5)}+\dfrac{2z+2}{(z-5)(z+1)}$

$\boxed{\dfrac{5z+2}{(z+1)(z-5)}}$

(18) $\dfrac{3}{x^2-4x-21}-\dfrac{1}{x^2-9x+14}$

$\dfrac{3}{(x+3)(x-7)}-\dfrac{1}{(x-2)(x-7)}$

$\dfrac{(x-2)}{(x-2)}\dfrac{3}{(x+3)(x-7)}-\dfrac{1}{(x-2)(x-7)}\dfrac{(x+3)}{(x+3)}$

$\dfrac{3x-6}{(x-2)(x+3)(x-7)}-\dfrac{x+3}{(x-2)(x-7)(x+3)}$

$\boxed{\dfrac{2x-9}{(x-2)(x+3)(x-7)}}$

(19) $\dfrac{y}{2y-1}+\dfrac{6}{y+4}$

$\dfrac{(y+4)}{(y+4)}\dfrac{y}{(2y-1)}+\dfrac{6}{(y+4)}\dfrac{(2y-1)}{(2y-1)}$

$\dfrac{y^2+4y}{(y+4)(2y-1)}+\dfrac{12y-6}{(y+4)(2y-1)}$

$\boxed{\dfrac{y^2+16y-6}{(y+4)(2y-1)}}$

(20) $\dfrac{3}{z}-\dfrac{2}{z^2-4z+3}$

$\dfrac{3}{z}-\dfrac{2}{(z-3)(z-1)}$

$\dfrac{(z-3)(z-1)}{(z-3)(z-1)}\dfrac{3}{z}-\dfrac{2}{(z-3)(z-1)}\dfrac{(z)}{(z)}$

$\dfrac{(z^2-4z+3)3}{z(z-3)(z-1)}-\dfrac{2z}{z(z-3)(z-1)}$

$\dfrac{3z^2-12z+9}{z(z-3)(z-1)}-\dfrac{2z}{z(z-3)(z-1)}$

$\boxed{\dfrac{3z^2-14z+9}{z(z-3)(z-1)}}$

(21) $\dfrac{3y-5}{y^2-2y+1}-\dfrac{4}{y}$

$\dfrac{3y-5}{(y-1)(y-1)}-\dfrac{4}{y}$

$\dfrac{(y)}{(y)}\dfrac{(3y-5)}{(y-1)(y-1)}-\dfrac{4}{y}\dfrac{(y-1)(y-1)}{(y-1)(y-1)}$

$\dfrac{3y^2-5y}{y(y-1)(y-1)}-\dfrac{4(y^2-2y+1)}{y(y-1)(y-1)}$

$\dfrac{3y^2-5y}{y(y-1)(y-1)}-\dfrac{4y^2-8y+4}{y(y-1)(y-1)}$

$\boxed{\dfrac{-y^2+3y-4}{y(y-1)(y-1)}}$

(22) $\dfrac{x+5}{x-1}+\dfrac{2}{x+4}$

$\dfrac{(x+4)}{(x+4)}\dfrac{(x+5)}{(x-1)}+\dfrac{2}{(x+4)}\dfrac{(x-1)}{(x-1)}$

$\dfrac{x^2+9x+20}{(x+4)(x-1)}+\dfrac{2x-2}{(x+4)(x-1)}$

$\boxed{\dfrac{x^2+11x+18}{(x+4)(x-1)}}$

(23) $\dfrac{x+2}{5-x}+\dfrac{2}{x+9}$

$\dfrac{(x+9)}{(x+9)}\dfrac{(x+2)}{(5-x)}+\dfrac{2}{(x+9)}\dfrac{(5-x)}{(5-x)}$

$\dfrac{x^2+11x+18}{(x+9)(5-x)}+\dfrac{10-2x}{(x+9)(5-x)}$

$\boxed{\dfrac{x^2+9x+28}{(x-9)(5-x)}}$

(24) $\dfrac{2}{z^2-16}-\dfrac{3}{z^2+4z}$

$\dfrac{2}{(z-4)(z+4)}-\dfrac{3}{z(z+4)}$

$\dfrac{(z)}{(z)}\dfrac{2}{(z-4)(z+4)}-\dfrac{3}{z(z+4)}\dfrac{(z-4)}{(z-4)}$

$\dfrac{2z}{z(z-4)(z+4)}-\dfrac{3z-12}{z(z+4)(z-4)}$

$\boxed{\dfrac{-z+12}{z(z-4)(z+4)}}$

(25) $\dfrac{y+5}{3y^2+7y+2}+\dfrac{2}{y+2}$

$\dfrac{y+5}{(3y+1)(y+2)}+\dfrac{2}{y+2}$

$\dfrac{y+5}{(3y+1)(y+2)}+\dfrac{2}{(y+2)}\dfrac{(3y+1)}{(3y+1)}$

$\dfrac{y+5}{(3y+1)(y+2)}+\dfrac{6y+2}{(y+2)(3y+1)}$

$\boxed{\dfrac{7y+7}{(3y+1)(y+2)}}$

(26) $\dfrac{2}{z^2-4}-\dfrac{2z}{z+2}$

$\dfrac{2}{(z-2)(z+2)}-\dfrac{2z}{z+2}$

$\dfrac{2}{(z-2)(z+2)}-\dfrac{2z}{(z+2)}\dfrac{(z-2)}{(z-2)}$

$\dfrac{2}{(z-2)(z+2)}-\dfrac{2z^2-4z}{(z+2)(z-2)}$

$\boxed{\dfrac{-2z^2+4z+2}{(z+2)(z-2)}}$

(27) $\frac{y-3}{2y^2+7y+6}+\frac{1}{y+2}$

$$\frac{y-3}{(2y+3)(y+2)}+\frac{1}{y+2}$$

$$\frac{y-3}{(2y+3)(y+2)}+\frac{1(2y+3)}{(y+2)(2y+3)}$$

$$\frac{y-3}{(2y+3)(y+2)}+\frac{2y+3}{(y+2)(2y+3)}$$

$$\boxed{\frac{3y}{(2y+3)(y+2)}}$$

(28) $\frac{2}{4z^2-9}-\frac{2z}{2z-7}$

$$\frac{(2z-7)2}{(2z-7)(4z^2-9)}-\frac{2z(4z^2-9)}{(2z-7)(4z^2-9)}$$

$$\frac{4z-14}{(2z-7)(4z^2-9)}-\frac{8z^3-18z}{(2z-7)(4z^2-9)}$$

$$\boxed{\frac{-8z^3+22z-14}{(2z-7)(4z^2-9)}}$$

(29) $-2+\frac{1}{y}+\frac{y-5}{y+9}$

$$-\frac{2}{1}+\frac{1}{y}+\frac{y-5}{y+9}$$

$$\frac{y(y+9)(-2)}{y(y+9)1}+\frac{1(y+9)}{y(y+9)}+\frac{(y-5)(y)}{(y+9)(y)}$$

$$\frac{-2y^2-18y}{y(y+9)}+\frac{y+9}{y(y+9)}+\frac{y^2-5y}{y(y+9)}$$

$$\boxed{\frac{-y^2-22y+9}{y(y+9)}}$$

(30) $\frac{1}{2y}+3-\frac{y-3}{y+5}$

$$\frac{1}{2y}+\frac{3}{1}-\frac{y-3}{y+5}$$

$$\frac{(y+5)1}{(y+5)2y}+\frac{2y(y+5)3}{2y(y+5)1}-\frac{(y-3)(2y)}{(y+5)(2y)}$$

$$\frac{y+5}{2y(y+5)}+\frac{6y^2+30y}{2y(y+5)}-\frac{2y^2-6y}{2y(y+5)}$$

$$\boxed{\frac{4y^2+37y+5}{2y(y+5)}}$$

(31) $\frac{2}{5x+1}+\frac{x-7}{25x^2-1}$

$$\frac{2}{5x+1}+\frac{x-7}{(5x-1)(5x+1)}$$

$$\frac{(5x-1)2}{(5x-1)5x+1}+\frac{x-7}{(5x-1)(5x+1)}$$

$$\frac{10x-2}{(5x-1)(5x+1)}+\frac{x-7}{(5x-1)(5x+1)}$$

$$\boxed{\frac{11x-9}{(5x-1)(5x+1)}}$$

(32) $\frac{-4}{4z^3-16z^2}+\frac{2z-3}{5z}$

$$\frac{-4}{4z^2(z-4)}+\frac{2z-3}{5z}$$

$$\frac{(5)(-4)}{(5)4z^2(z-4)}+\frac{(2z-3)4z(z-4)}{5z\,4z(z-4)}$$

$$\frac{-20}{20z^2(z-4)}+\frac{4z(2z^2-11z+12)}{20z^2(z-4)}$$

$$\frac{-20}{20z^2(z-4)}+\frac{8z^3-44z^2+48z}{20z^2(z-4)}$$

$$\boxed{\frac{8z^3-44z^2+48z-20}{20z^2(z-4)}}$$

(33) $\frac{1}{18x^2-45x}-\frac{3x}{5-2x}$

$$\frac{1}{9x(2x-5)}-\frac{3x}{5-2x}$$

$$\frac{1}{9x(2x-5)}-\frac{3x}{-1(-5+2x)}$$

$$\frac{1}{9x(2x-5)}-\frac{-3x(9x)}{(-5+2x)(9x)}$$

$$\frac{1}{9x(2x-5)}-\frac{-27x^2}{9x(-5+2x)}$$

$$\boxed{\frac{27x^2+1}{9x(2x-5)}}$$

(34) $\frac{-1}{4-3x}+\frac{x-7}{9x^2-16}$

$$\frac{-1}{4-3x}+\frac{x-7}{(3x-4)(3x+4)}$$

$$\frac{-1}{-1(-4+3x)}+\frac{x-7}{(3x-4)(3x+4)}$$

$$\frac{(3x+4)1}{(3x+4)(-4+3x)}+\frac{x-7}{(3x-4)(3x+4)}$$

$$\frac{3x+4}{(3x+4)(-4+3x)}+\frac{x-7}{(3x-4)(3x+4)}$$

$$\boxed{\frac{4x-3}{(3x+4)(3x-4)}}$$

(35) $\frac{1}{5x^3-45x}-\frac{-3x}{2x^3+2x^2-12x}$

$$\frac{1}{5x(x^2-9)}-\frac{-3x}{2x(x^2+x-6)}$$

$$\frac{2(x-2)1}{2(x-2)5x(x-3)(x+3)}-\frac{-3x\,5(x-3)}{2x(x+3)(x-2)5(x-3)}$$

$$\frac{2x-4}{10x(x-2)(x-3)(x+3)}-\frac{-15x(x-3)}{10x(x+3)(x-2)(x-3)}$$

$$\frac{2x-4}{10x(x-2)(x-3)(x+3)}-\frac{-15x^2+45x}{10x(x+3)(x-2)(x-3)}$$

$$\boxed{\frac{15x^2-43x-4}{10x(x-2)(x-3)(x+3)}}$$

(36) $\frac{-4}{3z^3+33z^2+72z}+\frac{6}{z^2+3z}$

$$\frac{-4}{3z(z^2+11z+24)}+\frac{6}{z(z+3)}$$

$$\frac{-4}{3z(z+3)(z+8)}+\frac{6}{z(z+3)}$$

$$\frac{-4}{3z(z+3)(z+8)}+\frac{6\cdot 3(z+8)}{z(z+3)3(z+8)}$$

$$\frac{-4}{3z(z+3)(z+8)}+\frac{18(z+8)}{3z(z+3)(z+8)}$$

$$\frac{-4}{3z(z+3)(z+8)}+\frac{18z+144}{3z(z+3)(z+8)}$$

$$\boxed{\frac{18z+140}{3z(z+3)(z+8)}}$$